HOW ARE PARENTAL EDUCATION LEVELS ASSOCIATED WITH TRADITIONAL GENDER-ROLE ATTITUDES?

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Recap of Presentation 1

Examining the association between Parental Education Levels and Traditional gender-role attitudes

Alternative Hypotheses:

H1a - Mother's education level is associated with traditional gender-role attitudes ($r \neq 0$)

H1b - Father's Education level is associated with traditional gender-role attitudes ($r \neq 0$)



What we have since worked on

Expanded on our research scope, to assess how mothers' and fathers' education level compare in how they affect traditional gender-role attitudes

Previous Alternative Hypotheses:

- **Mother's education level is associated with traditional gender-role attitudes ($r \neq 0$)**
- **Father's education level is associated with traditional gender-role attitudes ($r \neq 0$)**

New Alternative Hypotheses:

- **Higher maternal education levels as compared to paternal education levels result in higher egalitarian gender attitudes ($r \neq 0$)**
- **Higher paternal education levels as compared to maternal education levels result in higher egalitarian gender attitudes ($r \neq 0$)**

Previous Null Hypotheses:

- **Mother's education level is not associated with traditional gender-role attitudes ($r = 0$)**
- **Father's education level is not associated with traditional gender-role attitudes ($r = 0$)**

New Null Hypotheses:

- **Higher maternal education levels as compared to paternal education levels does not result in higher egalitarian gender attitudes ($r = 0$)**
- **Higher paternal education levels as compared to maternal education levels does not result in higher egalitarian gender attitudes ($r = 0$)**

What we have since worked on

Researched on specific theories that tie into our research focus

Socialisation theory

- Too general of a concept
- Lack of a clear established link between the concept and our research focus

- Bourdieu's Cultural Reproduction
  - Parental education shaping family socialisation → Shapes individual disposition, gender-role attitudes included
- Bronfenbrenner's Ecological Theory
  - Formation of individual gender roles also dependent on the micro-environmental system in contact with an individual
  - Parents as one of the most direct microsystems

What we have since worked on

Ran our data through other methods of analysis

- Cronbach's Alpha
- Spearman's R Correlation
- Principal Component Analysis (New)
- OLS Regression (New)

Explored other potential moderating factors which affect formation of traditional gender role attitudes

Conducted Multivariate analysis with the following moderating variables

- Religiosity
- Political Views
- Attitudes toward technology
- Demographic Info

Education and Gender Attitudes

Education and women's economic empowerment (Garrido, 2022)

- Education encourages more tolerant gender attitudes.
- Improves women's economic opportunities.

Higher education and support for gender equality in Singapore (Matthews, 2021)

- Higher education associated with stronger support for gender equality.
- Younger and more affluent respondents were more egalitarian in Singapore.

Parental Influence and Socialisation

Parental education and family resources (Buchmann et al, 2008)

- Parental education influences children's development.
- Family resources shape developmental opportunities.

Early exposure to traditional domestic roles (Matthews, 2021)

- Early exposure to domestic labour divisions.
- Traditional household roles shape gender expectations.

Theoretical Framework: Cultural Capital (Bourdieu, 1986)

Three Forms of Cultural Capital

- Embodied: Values, dispositions and ways of thinking.
- Objectified: Cultural goods and resources.
- Institutionalised: Educational qualifications.

Intergenerational Influence

- Family background influences children's cultural exposure.
- Parental experiences and values may be passed across generations.

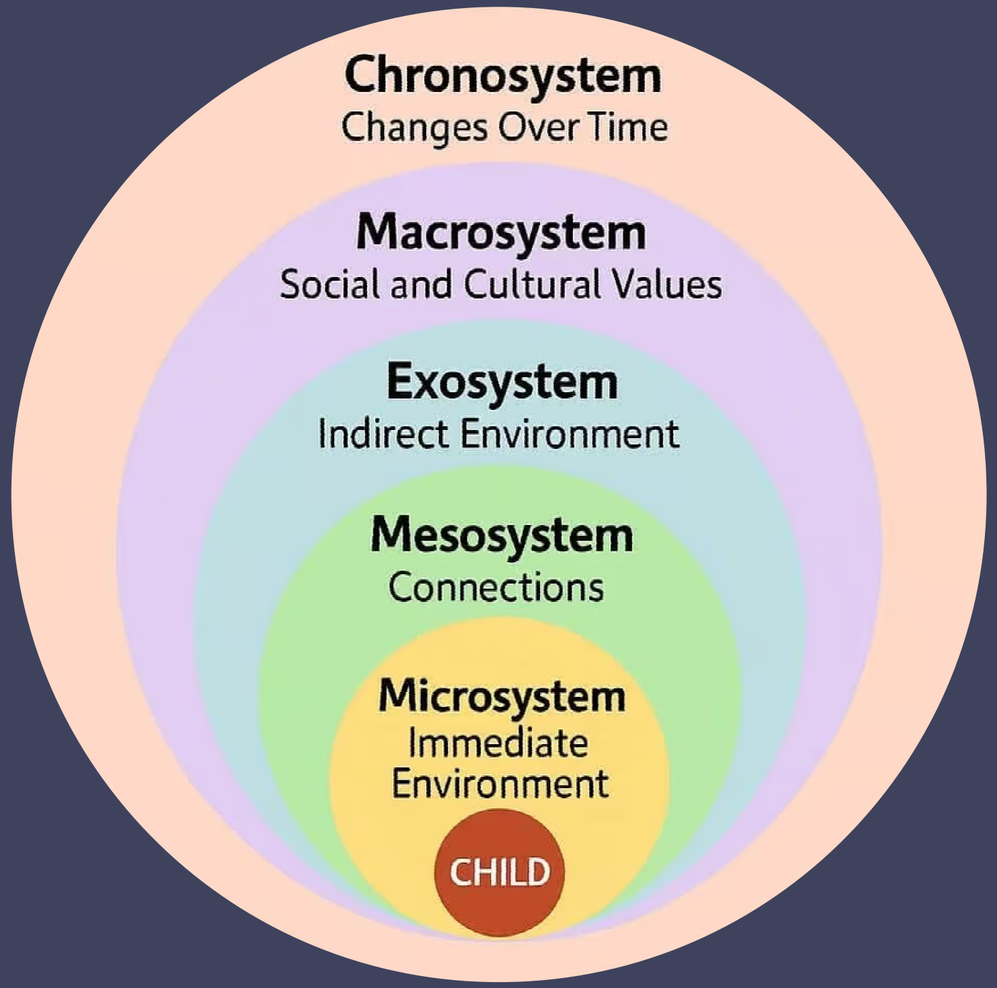
Family Transmission

- Cultural values acquired through family socialisation.
- Everyday interactions shape children's dispositions.

Relevance to Our Study

- Parental education may shape family values and socialisation.
- Family socialisation may influence individual gender-role attitudes.
- Higher parental education does not necessarily mean more egalitarian views.

Theoretical Framework: Ecological Theory (Bronfenbrenner, 1979)



- Human development (in our thoughts, attitudes, and behaviour) is shaped by a series of interconnected environmental systems

- **Microsystems**
  - Bi-directional relationships: Just as a person affects others, the immediate environment shapes the individual
- **Exosystems**
  - Contexts that the individual does not directly participate in but are still indirectly affected by



Relevance to Our Study

- Families, and by extension parents are one of the first, most direct microsystems an individual comes into contact with
- Traditional gender-role attitudes as influenced by parents(microsystem), whose perspective may themselves be informed by their own educational backgrounds and outcomes (exosystem)

Methods

Cleaned final variables

Re-cleaned gender-role, parental education, religiosity, science/technology, political-culture, age, sex and own education

Mother>Father

Grouped respondents into three parental-education profiles and compared their traditional gender-role attitude scores.

Moderation

Tested whether religiosity, science/technology attitudes and political culture changed the relationship between parental education and gender-role attitudes.

Exploration beyond RQ

Examined how respondents' own age, sex and education relate to gender-role attitudes

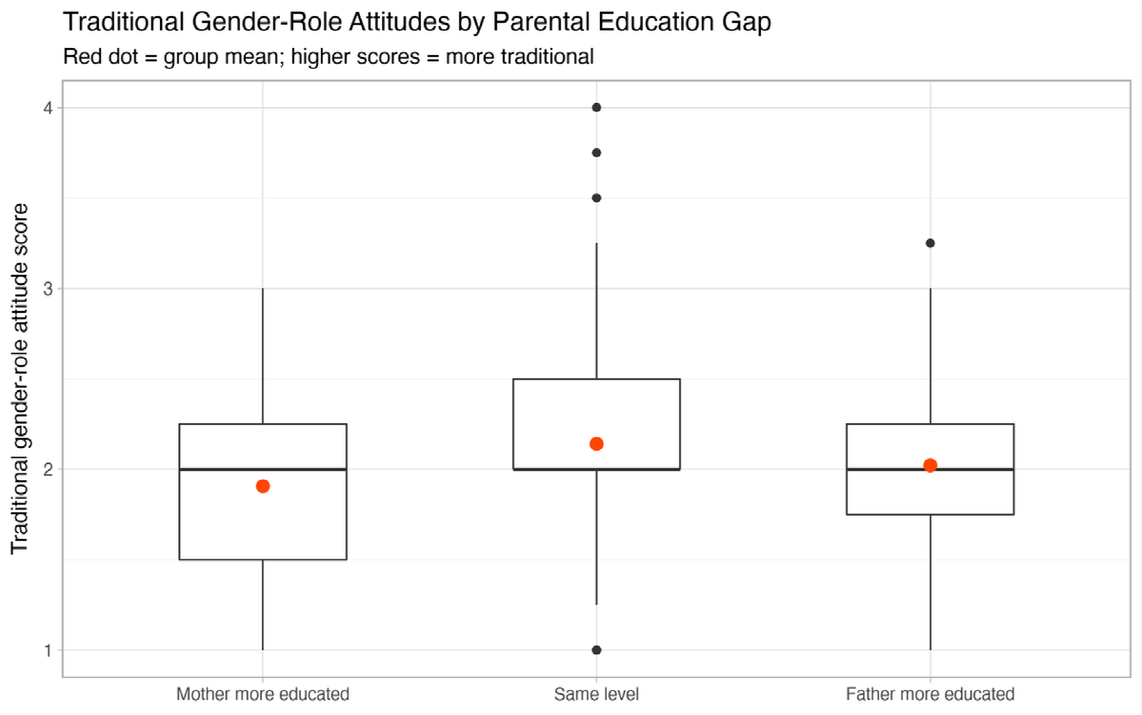
PCA

Explored how parental education, religiosity, science and political attitudes, and gender-role attitudes vary together

Multivariate Models

Modelled maternal and paternal education together, then added moderating variables, sex, age and respondents' own education

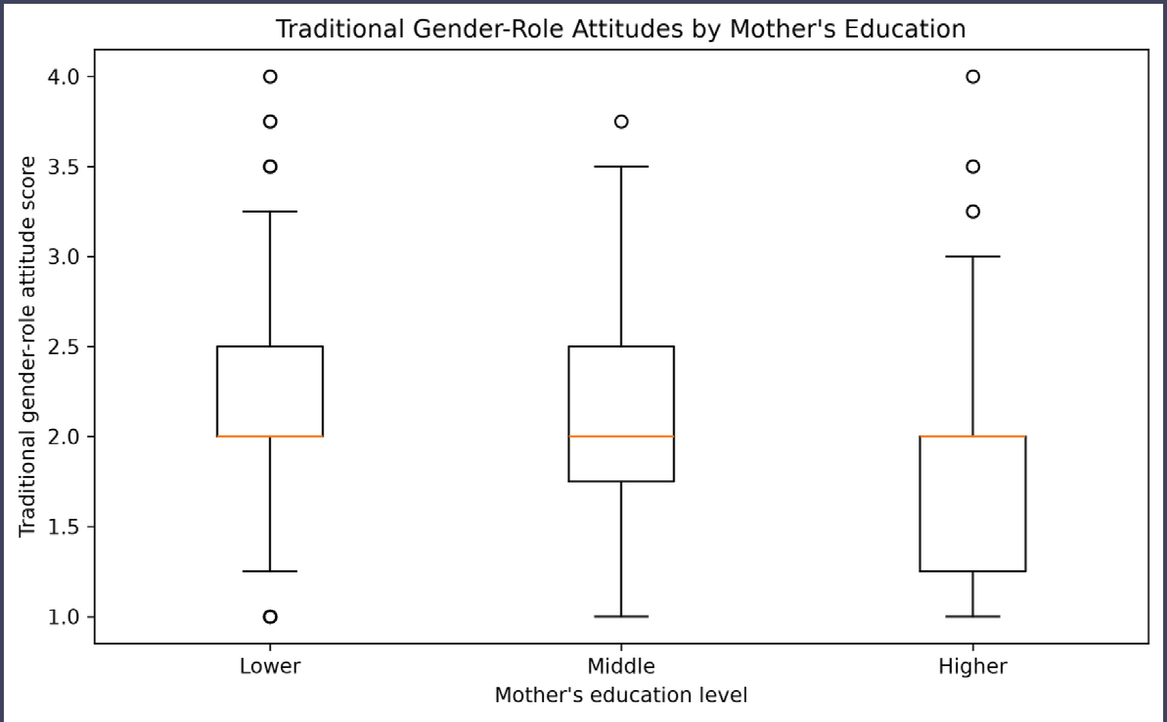
Mother > Father



- Grouped participants into 3 groups: mother more educated, same level, or father more educated.
- The mother-more-educated group seems to be the least traditional but it does not differ significantly from the father-more-educated group.
- From Welch's t-test and Cohen's d:
  - **Mother > Father**: least traditional attitudes (M = 1.91), significantly lower than same-level parents (M = 2.14, p < .001, d = -0.42).
  - **Father > Mother**: also less traditional than same-level parents, but with a smaller effect (p < .001, d = -0.21).
  - Mother more educated vs Father more educated: No significant difference

Comparison	Mean difference	Welch's t	p-value	Cohen's d
Mother more educated vs Same level	-0.23404304029304000	-4.295355560497930	3.55270534377696E-05	-0.42385754850251300
Father more educated vs Same level	-0.11891146134567200	-3.359387004882460	0.0008483951464734620	-0.2144340679011260
Mother more educated vs Father more educated	-0.11513157894736900	-1.874369738962620	0.062449058204916800	-0.20821465932365600

Is the child less traditional because the mother is more educated than the father, or simply because the mother herself is more highly educated?



- Mother's education level: traditional gender-role attitudes became less traditional as maternal education increased
- To see whether the difference was really due to the mother being more educated than the father, we compared families where the mothers had the same education level.
- Once maternal education was held constant, the Mother > Father differences became small and non-significant:
  - Middle education: Δ Mean = -0.09, p = .221, d = -0.17
  - Higher education: Δ Mean = -0.08, p = .437, d = -0.12

Mother's own education appears more important than whether she is more educated than the father.

Mother's education	Comparison	n (a)	n (b)	Mean difference	Welch's t	p-value	Cohen's d
Middle	Mother more educated vs Same level	63	236	-0.09401062684960990	-1.2314919124209400	0.2209041587162140	-0.16647762134290200
Higher	Mother more educated vs Same level	41	127	-0.07816400998655660	-0.7801714554347100	0.4374447177859230	-0.12391077102770300

Moderation

Tested whether religiosity, science attitudes and political views change the parental education link.

Following three interactions are the only ones that held up under every check (both parents in the same model, parental education coded as categories etc.)

Parent × Moderator	Interaction (b3)	p	Extra variance explained
Mother × Liberal notions of democracy	-0.044	< .001	0.9%
Mother × Authoritarian notions of democracy	0.041	< .001	0.7%
Father × Science & faith/morality	-0.034	< .001	0.8%

Mother x Liberal

Moderator: higher score means a more liberal view of democracy

Mother's education was more strongly linked to less traditional attitudes among people with more liberal views of democracy.

Mother x Authoritarian

Moderator: higher score means seeing authoritarian features as part of democracy

Mother's education was barely linked to attitudes among people who see authoritarian features as part of democracy.

Father x Science & faith/morality

Moderator: higher score means less worried that science threatens faith and morality

Father's education was more strongly linked to less traditional attitudes among people less worried that science threatens faith and morality.

Note: These moderation effects were statistically reliable, but small (each explained less than 1% additional variance)

Multivariate models

	Model 1	Model 2	Model 3	Model 4
Added	Parents' education	+ Mother more educated	+ Values	+ Sex, age, own education
Mother's education	-0.076*	-0.011	0.020	0.061
Father's education	-0.071*	-0.116*	-0.099*	-0.029
Mother more educated	–	-0.217*	-0.178*	-0.113
R^2	3.0%	3.6%	15.3%	24.6%

- **Built four regression models step-by-step**: starting with maternal and paternal education, then adding the parental education gap, respondents' values, and finally sex, age and own education.

- Parental education was associated with gender-role attitudes in the simpler models, but these associations weakened as other factors were added.
  - In the final model, mother's education, father's education and the parental education gap were no longer statistically significant.

- The final model explained about 25% of the variation in gender-role attitudes (Best Model).
  - The largest associations in the final model were with authoritarian views, age, sex, science & faith/morality attitudes and respondents' own education.

At first, more educated parents were linked to less traditional attitudes. However, once we considered respondents' age, sex, own education and values, parents' education was no longer a significant predictor.

Exploration Beyond RQ

In our multivariate model, age, sex and respondents' own education remained significant predictors while parental education did not. We therefore explored these factors further.

Characteristic	ρ with attitudes
Age	0.307
Own education	-0.277
Female	-0.130
Mother's education (Presentation 1)	-0.171
Father's education (Presentation 1)	-0.160

- Note: $p < .001$ for all

Older respondents were more traditional, while women and more highly educated respondents were less traditional.

- Age: $\rho = .31$
- Own education: $\rho = -.28$
- Female: $\rho = -.13$

Age and respondents' own education were more strongly related to gender-role attitudes than either parent's education.

Interpretations & Limitations

Interpretations

Our study suggests that parental education shapes part of the story, but not the whole story.

- The mother's education appears more important than whether she out-educated the father.
- The influence of parental education depends on the wider social context.

Overall, gender-role attitudes appear to reflect both family background and respondents' own social development.

- Fits an ecological view

Limitations

- Associations cannot establish causality or show how attitudes change over time.
- Parental education was limited to formal qualifications, potentially missing informal learning, certifications and other educational experiences.
- Small Key Group: Only 104 respondents with a more educated mother, so low power to detect a modest gap effect.

Future Directions:

- **Look beyond parents' education levels by measuring their actual gender attitudes and how responsibilities were divided at home.**
- **Study influences outside the family, such as friends, partners, schools and workplaces, to understand how gender-role attitudes continue to develop over time.**

THANK YOU FOR LISTENING